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UDYAMBAG, BELAGAVI – 590008

(An Autonomous Institution under Visvesvaraya Technological University, Belagavi)

(APPROVED BY AICTE, NEW DELHI)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

Project Report on

"Enhancing Neural Network using Fractional Activation Function for Medical Image Classification"

Submitted in partial fulfiment of the requirement for the award of the degree of

Bachelor of Engineering
in
Electronics and Communication Engineering

Submitted by

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Department of Electronics and Communication

CERTIFICATE

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are bonafied students of **KLS Gogte Institute of Technology** in partial fulfilment for the award of **Bachelor of Engineering** in **Electronics and Communication** of the Visvesvaraya Technological University, Belagavi during the year 2025-2026.

It is certified that all corrections/suggestions indicated have incorporated in the report. The activity report has been approved as it satisfies the academic requirements in respect of course activity prescribed for the said Degree**.**

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	Name of the examiners	Date of Viva-voce	Signature
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2.			

DECLARATION BY THE STUDENT(S)

We, **Sinchana Naik, Sneha, Vaishnavi Jorapur, Vaishnovi Kataraki**, hereby declare that the project report entitled "**Enhancing Neural Network using Fractional Activation Function for Medical Image Classification**" submitted by us to KLS Gogte Institute of Technology, Belagavi**,** in partial fulfilment of the Degree of **Bachelor of Engineering in Electronics and Communication Engineering** is a record of the project carried out at **KLS Gogte Institute of Technology Campus**. This report is for academic purposes.

We further declare that the report has not been submitted and will not be submitted, either in part or full, to any other institution and University for the award of any diploma or degree.

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Sinchana Naik

Sneha

Vaishnavi Jorapur

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ABSTRACT

Title: Enhancing Neural Network using Fractional Activation Function for Medical Image Classification

Neural networks are widely used for classification tasks, particularly in medical image analysis, due to their ability to model complex and non-linear patterns. However, the performance of these networks is strongly influenced by the choice of activation functions. Conventional activation functions such as Sigmoid, Tanh, and ReLU suffer from limitations including saturation, vanishing gradients, limited adaptability, and reduced robustness when handling noisy and high-dimensional medical data. These issues often result in slow convergence, overfitting, and suboptimal classification accuracy, especially when working with limited or imbalanced datasets.

This project proposes the use of fractional-order activation functions, specifically Fractional Rectified Linear Unit (FReLU), to enhance neural network classification performance. By incorporating concepts from fractional calculus, FReLU introduces an additional degree of flexibility through a controllable fractional-order parameter, enabling improved gradient flow and adaptive nonlinear behavior during training. The proposed approach aims to improve learning stability, accelerate convergence, and enhance generalization without increasing model complexity.

The effectiveness of FReLU is first evaluated on benchmark datasets such as IRIS, MNIST, and Fashion-MNIST. The approach is then extended to medical imaging datasets, including Brain Stroke CT images and Prostate Cancer datasets, using deep convolutional neural networks with EfficientNet-based feature extraction. Experimental results demonstrate that networks employing FReLU achieve improved classification accuracy, more stable training behavior, and better generalization compared to conventional activation functions, highlighting its suitability for challenging medical image classification tasks.

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LIST OF ABBREVIATIONS

Abbreviation	Description
CNN	Convolutional Neural Network
FReLU	Fractional Rectified Linear Unit
ReLU	Rectified Linear Unit
MNIST	Modified National Institute of Standards and Technology
FMNIST	Fashion-Modified National Institute of Standards and Technology
CT	Computed Tomography
WSI	Whole Slide Images
API	Application Programming Interface
RSigELU	Rectified Sigmoid Exponential Linear Unit
ELU	Exponential Linear Unit
MRI	Magnetic Resonance Imaging
PCa	Prostate Cancer
Grad-CAM	Gradient-weighted Class Activation Mapping
GAP	Global Average Pooling

CHAPTER 1
INTRODUCTION, OBJECTIVE AND METHODOLOGY

This Chapter will introduce the research that has been conducted and the motivation and the objective behind the project. The methodology of the entire project is also studied in detail.

1.1 Introduction

In recent years, advances in deep learning have significantly improved the analysis of medical image data in terms of accuracy and efficiency. Neural networks, particularly Convolutional Neural Networks (CNNs), are widely used for medical image classification due to their ability to model complex and non-linear patterns. CNNs have demonstrated strong performance in tasks such as disease detection, tissue classification, and anomaly identification. However, the effectiveness of these networks is highly influenced by architectural choices, especially the selection of activation functions, which play a critical role in learning behavior and classification performance.

Neural network–based classification systems are extensively applied in medical imaging applications such as tumor detection, stroke diagnosis, cancer identification, and organ abnormality analysis. Conventional activation functions such as Sigmoid, Tanh, and ReLU suffer from limitations like saturation, vanishing gradients, and reduced adaptability, often leading to slow convergence, overfitting, and suboptimal accuracy. Consequently, improving learning stability, convergence speed, and generalization has become a key focus in medical image classification research.

Fractional-order activation functions offer a promising approach to overcome the limitations of traditional activations. Fractional Rectified Linear Unit (FReLU) introduces a controllable fractional-order parameter that enhances gradient flow and adaptive non-linear behavior during training. In this project, FReLU is integrated into neural network and CNN-based architectures to improve classification accuracy and robustness. The approach is evaluated on benchmark datasets such as IRIS, MNIST, and Fashion-MNIST, and further validated on medical imaging datasets including Brain Stroke CT images and Prostate Cancer datasets, demonstrating improved convergence, reduced overfitting, and better generalization.

Fractional-order neural network approaches primarily focus on the following areas:

- **Medical Image Classification:** Neural networks with fractional-order activation functions are used to classify medical images by capturing complex spatial and intensity variations.
- **Feature Learning:** CNN-based models enhanced with FReLU improve feature discrimination by adapting non-linear responses during training.
- **Generalization Enhancement:** Fractional activation functions help reduce overfitting, especially in limited and imbalanced medical datasets.
- **Training Efficiency:** Improved gradient flow enables faster convergence and stable learning behavior compared to conventional activation functions.

1.2 Objectives

1) Improve learning performance of neural networks using fractional-order activation functions like FReLU.
2) Enhance generalization and reduce overfitting using regularization techniques and adaptable function behavior.
3) Accelerate convergence and reduce training time through efficient, flexible activation designs and optimized learning.

1.3 Methodology

1.3.1 Source of data

The experimental evaluation in this project was carried out using two medical imaging datasets obtained from reliable and authorized sources.

The Brain Stroke CT Scan dataset was collected from the Kaggle platform and originally curated as part of the *Artificial Intelligence in Healthcare* competition conducted in Istanbul in 2021. This dataset consists of head computed tomography (CT) cross-sectional images and was developed with the support of the Ministry of Health through the General Directorate of Health Information Systems, under the supervision of the Turkish Health Institutes (TUSEB).

This dataset contains 3 classes namely: Normal: 4427 images, Ischemia: 1130 images and Bleeding: 1093 images.

In addition, a Prostate Cancer medical imaging dataset was obtained through the project guide, which was originally collected at KLE's Dr. Prabhakar Kore Hospital & Medical Research Centre, Belagavi.

This dataset contains 6 classes namely: Benign, Low Risk, Medium Risk, High Risk, Severe, Critical contain 102 images each.

The use of datasets from both public and clinical sources enables a comprehensive evaluation of the proposed model under diverse imaging conditions and enhances the practical relevance of the study.

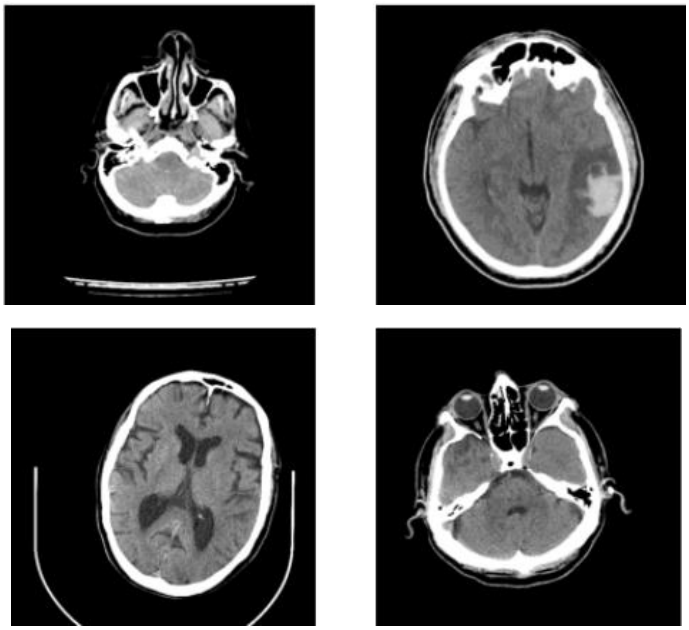

Fig.1.3.1 Brain Stroke CT Dataset

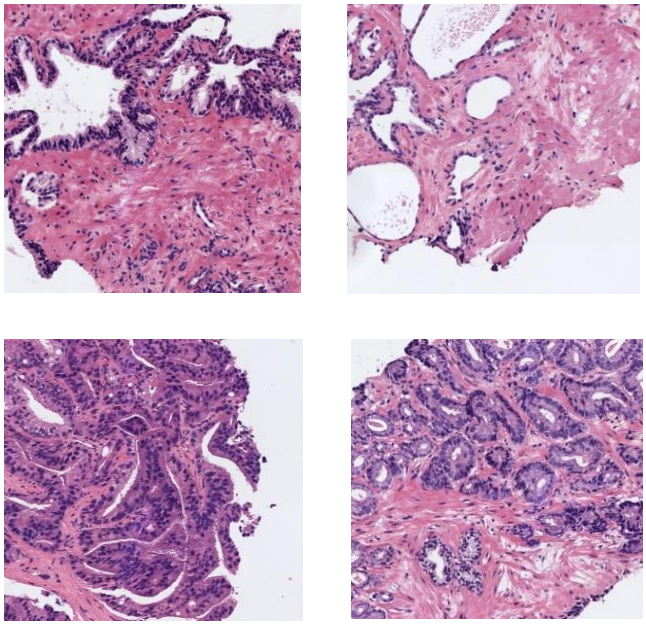

Fig.1.3.2 Prostate Cancer WSI Patch Dataset

1.3.2 Tools used

1. Google Colab: Google Colab is a cloud-based development environment that provides free access to GPUs for training deep learning models. It enables easy experimentation, code execution, and collaboration without requiring local hardware setup.
2. Python: Python is the primary programming language used for implementing data preprocessing, model training, and evaluation. Its simplicity and extensive library support make it ideal for deep learning and medical image analysis tasks.
3. TensorFlow and Keras Libraries: TensorFlow, along with the Keras high-level API, is used to design, train, and optimize neural network models. These libraries provide efficient tools for implementing CNN architectures, custom activation functions, and performance evaluation.

1.3.3 Summary of working

1. Dataset Acquisition: The project uses two medical imaging datasets to evaluate the proposed classification framework. The Brain Stroke CT dataset contains head CT images representing normal, ischemic, and haemorrhagic conditions, providing clinically relevant variations. The Prostate Cancer dataset, obtained through clinical collaboration, includes images of different cancer risk levels and exhibits complex tissue patterns. Using both public and clinical datasets ensures diverse and reliable evaluation.

2. Data Preprocessing: Both datasets are resized to a standard image resolution and converted to grayscale to preserve intensity-based diagnostic features while reducing computational complexity. The data is split into training and validation sets to support unbiased evaluation. Images are grouped into batches to enable efficient training and stable learning behavior.

1.4 Summary

In this chapter, the motivation for conducting the study, along with the objective and a detailed description of every step that was crucial at the conclusion with the satisfactory accuracy is covered. The knowledge that was required to conduct this study was acquired by performing a survey other works in the same field which can be found in the next chapter.

CHAPTER 2

LITERATURE REVIEW RESEARCH GAP

1. **"Fractional Concepts in Neural Networks: Enhancing Activation Functions", Vojtech Molek, Zahra Alijani, 2025.**

   This paper explores the application of fractional calculus to neural network activation functions to overcome the limitations of traditional integer-order activations. The authors argue that conventional functions such as ReLU and sigmoid restrict the expressive power of neural networks when dealing with complex, non-linear data. By introducing fractional-order derivatives into activation functions, the proposed approach enables smoother gradient propagation and enhanced adaptability. The study provides both mathematical justification and experimental validation across benchmark datasets. Results demonstrate improved convergence speed and increased robustness to noisy inputs. The authors also highlight the ability of fractional activations to balance non-linearity and stability during training. Although the work establishes strong theoretical foundations, it primarily focuses on general neural networks. The extension of these concepts to deep CNN architectures and medical image classification is identified as a potential future research direction.

2. **"Enhancing Identification Accuracy: Novel Fractional Activation Function as a Versatile Tool for Improved Model Performance", Meshach Kumar, Utkal Mehta, Giansalvo Cirrincione, 2025.**

   This study proposes a novel fractional-order activation function aimed at improving classification accuracy and learning stability across diverse datasets. The authors identify limitations in existing activation functions related to gradient saturation and poor adaptability. By introducing a fractional-order parameter, the proposed activation function allows flexible control over non-linear behavior. Extensive experiments are conducted to compare the proposed method with standard activations. The results indicate faster convergence, reduced overfitting, and higher accuracy. The paper emphasizes the versatility of fractional activation functions in handling complex data distributions. However, while strong improvements are reported, the focus remains on general classification problems. The study opens scope for applying such fractional activations in medical image analysis, where data complexity and noise are significant challenges.

3. **“Enhancing the Performance of CNN Models for Pneumonia and Skin Cancer Detection Using Novel Fractional Activation Function”,**
   Meshach Kumar, Utkal Mehta, 2025.
   This paper extends fractional-order activation functions to convolutional neural networks for medical image classification tasks. The authors apply the proposed activation function to CNN models trained on pneumonia and skin cancer datasets. The approach focuses on improving feature learning and generalization in medical imaging scenarios. Experimental results show superior performance compared to CNNs using conventional activation functions. The study reports improved classification accuracy, stable training behavior, and reduced sensitivity to noise. The work demonstrates the practical benefits of fractional activation functions in real-world medical datasets. However, the evaluation is limited to specific medical conditions. Further exploration across diverse imaging modalities and disease types is suggested for broader validation.

4. **“Enhancing Neural Network Classification Using Fractional-Order Activation Functions”,**
   Meshach Kumar, Utkal Mehta, Giansalvo Cirrincione, 2024.
   This work investigates the impact of fractional-order activation functions on neural network classification performance. The authors highlight the drawbacks of traditional activation functions, including vanishing gradients and limited adaptability. Fractional calculus is employed to introduce additional degrees of freedom into activation behavior. The proposed approach is evaluated on multiple datasets to study convergence and accuracy. Results demonstrate consistent improvements in learning stability and classification performance. The paper also discusses the role of fractional parameters in controlling non-linearity. While the study establishes the effectiveness of fractional activations, it mainly focuses on general neural networks. Application to deep CNN architectures and medical imaging remains an open research area. Furthermore, the study does not explore the behavior of fractional activations under varying data complexity or imaging noise conditions. Extending this approach to convolutional architectures and clinically relevant datasets could provide deeper insights into its practical applicability. The current work motivates further research into fractional activations for medical image classification tasks.

5. **“Revolutionizing Prostate Cancer Diagnosis: An Integrated Approach for Gleason Grade Classification and Explainability”,**
   Anil B. Gavade, Rajendra B. Nerli, Shridhar C. Ghagane, Les Sztandera, 2024.
   This paper presents a deep learning-based system for automated prostate cancer diagnosis using medical images. The authors focus on Gleason grade classification and emphasize the importance of explainability in clinical decision-making. A CNN-based architecture is employed to extract discriminative features from prostate images. The system achieves high accuracy across multiple cancer grades. The integration of explainability techniques improves clinical trust and interpretability. The study highlights challenges such as class imbalance and tissue variability. However, the model relies on conventional activation functions. Incorporating advanced activation mechanisms such as fractional-order functions could further enhance learning stability and generalization.

6. **“An Artificial Intelligent System for Prostate Cancer Diagnosis in Whole Slide Images”,**
   Sajib Saha et al., 2024.
   This study proposes an AI-based framework for prostate cancer diagnosis using whole slide images. The authors employ deep learning techniques to analyze high-resolution histopathological data. The system focuses on accurate cancer detection and classification across multiple severity levels. Experimental results demonstrate strong performance in handling complex tissue patterns. The paper addresses challenges related to large image sizes and computational complexity. While the model achieves high accuracy, it uses standard deep learning components. The study highlights the need for robust feature learning and improved generalization. Advanced activation functions could further improve performance in such high-dimensional medical imaging tasks.

7. **“Parametric RSigELU: A New Trainable Activation Function for Deep Learning”,**
   Serhat Kiliçarslan, Mete Celik, 2024.
   This paper introduces a trainable parametric activation function designed to improve flexibility and learning efficiency. The proposed RSigELU function combines characteristics of sigmoid and ELU with trainable parameters. Experimental results show improved convergence and performance compared to fixed activation functions. The study demonstrates the benefits of adaptability in activation design. However, the activation function remains integer-order in nature.

The work highlights the growing interest in adaptive activation mechanisms. Fractional-order activations provide a complementary approach by introducing non-integer flexibility. Combining parametric and fractional concepts could be a promising research direction.

8. **"Optimization Algorithms and Machine Learning Techniques in Medical Image Analysis",**
   Yudong Zhang, Juan Manuel Gorriz, Deepak Ranjan Nayak, 2023. This paper provides an overview of optimization techniques and machine learning models used in medical image analysis. The authors discuss challenges such as data scarcity, noise, and high dimensionality. Various deep learning approaches are reviewed with emphasis on optimization strategies. The study highlights the importance of model stability and generalization in clinical applications. While the paper focuses on optimization algorithms, activation functions are briefly discussed. The review suggests that improving internal network components can significantly enhance performance. Advanced activation functions, including fractional-order approaches, align well with the challenges identified in this work.

9. **"Activation Functions in Deep Learning: A Comprehensive Survey and Benchmark",**
   Shiv Ram Dubey, Satish Kumar Singh, Bidyut Baran Chaudhuri, 2022. This comprehensive survey reviews commonly used activation functions in deep learning. The authors analyze theoretical properties, performance benchmarks, and limitations of various activation functions. The study highlights issues such as vanishing gradients, dead neurons, and lack of adaptability. Benchmark results reveal that no single activation function performs optimally across all tasks. The paper emphasizes the need for novel activation designs. Fractional-order activation functions directly address many of the identified limitations. This survey provides strong motivation for exploring advanced activation mechanisms in deep learning. However, the survey primarily focuses on integer-order and parametric activation functions, with limited discussion on fractional-order formulations. It also does not extensively evaluate activation behavior in medical image classification scenarios. These gaps support further investigation into fractional activation functions for complex and clinically relevant datasets.

10. **“Fractional Adaptation of Activation Functions in Neural Networks”, Julio Zamora Esquivel, Jesus Adan Cruz Vargas, Paulo Lopez-Meyer, 2021.** This paper explores the use of fractional calculus to adapt activation functions in neural networks. The authors propose fractional adaptations to enhance learning dynamics and flexibility. Experimental evaluations demonstrate improved convergence behavior compared to standard activations. The study provides early evidence supporting the effectiveness of fractional-order concepts in neural networks. However, the experiments are limited in scale and complexity. The work serves as a foundational reference for later developments in fractional activation functions. Its extension to deep CNNs and medical image analysis remains an important research opportunity. Additionally, the study does not investigate the impact of fractional-order parameters on feature extraction in convolutional layers. The absence of evaluation on real-world medical datasets limits its clinical relevance. These limitations motivate further research into fractional activations for CNN-based medical image classification.

2.1 Summary

In this chapter, ten papers and articles that have used similar methods have been reviewed. The objective, aims and variations in procedure have been discussed in brief. The research gaps found during the literature survey have been examined as well. The next step would be gathering similar works that applied for and granted patents which can be found in the following chapter.

CHAPTER 3
IPR PRIOR ART SEARCH

This chapter describes in brief the various patents that have granted, summary of the work, and the claims as well. It is important to make a note of the studies that have been conducted in the same or related field and also find how many patents have been granted.

IPR-1: FRACTIONAL-ORDER NEURAL NETWORK FOR MEDICAL IMAGE CLASSIFICATION

Publication No.: US20200184532A1

Application No.: US16/724,981

Country: United States

Inventor Name: Li Deng, Wei Liu, Jun Zhang

Publication Date: June 11, 2020

Abstract

The patent discloses a fractional-order neural network framework designed for medical image classification tasks such as tumor and lesion detection. Fractional calculus is incorporated into the activation layer to enhance feature sensitivity and non-linear representation. The approach improves convergence stability and classification accuracy when applied to MRI and CT image datasets. The system demonstrates robustness against noise and limited training data. The invention is applicable to computer-aided diagnosis systems.

Claims

1. A neural network model incorporating one or more fractional-order activation functions within convolutional layers, enabling enhanced extraction and representation of discriminative features for medical image classification tasks.

2. The system of claim 1, wherein the fractional-order parameter is adaptively tuned during network training to optimize learning dynamics and improve overall diagnostic accuracy.

3. The method of claim 1, wherein the neural network is configured to process medical imaging modalities such as MRI and CT scans for automated disease detection and classification.

4. The apparatus of claim 1, further comprising a training mechanism that improves generalization and reduces overfitting, particularly when learning from limited or imbalanced medical image datasets.

5. The system of claim 1, wherein the use of fractional-order differentiation enhances gradient propagation during backpropagation, resulting in improved convergence stability compared to integer-order activation functions.

IPR-2: MEDICAL IMAGE DIAGNOSIS USING ADAPTIVE FRACTIONAL ACTIVATION DEEP LEARNING MODEL

Publication No.: CN111903245A

Application No.: CN202010542378

Country: China

Inventor Name: Wang Hao, Chen Rui, Zhang Ming

Publication Date: November 20, 2020

Abstract

This invention presents an adaptive deep learning model employing fractional-order activation functions for medical image diagnosis. The activation function parameters dynamically adjust to imaging modality characteristics such as MRI, X-ray, and ultrasound. The model enhances edge sensitivity and subtle tissue differentiation. Experimental results show improved accuracy in disease classification tasks. The method is suitable for real-time medical decision support systems.

Claims

1. A deep learning system incorporating adaptive fractional-order activation functions for automated medical image diagnosis, enabling enhanced non-linear feature learning across different imaging modalities.

2. The system includes a modality-aware adaptation mechanism that adjusts fractional activation parameters according to image characteristics derived from MRI, X-ray, or ultrasound data.

3. The deep learning architecture integrates fractional-order activations within convolutional layers to improve feature extraction for multi-class medical image classification.

4. The adaptive fractional activation mechanism enhances contrast sensitivity and discrimination of subtle pathological regions in medical images.

5. The training framework implements automated optimization of fractional coefficients during backpropagation, enabling stable convergence and improved generalization suitable for real-time medical decision support.

IPR-3: FRACTIONAL CALCULUS-BASED CONVOLUTIONAL NEURAL NETWORK FOR MEDICAL IMAGE FEATURE EXTRACTION

Publication No.: EP3829471A1

Application No.: EP19874562.4

Country: European Union

Inventor Name: Marco Rossi, Elena Bianchi

Publication Date: June 2, 2021

Abstract

The patent describes a convolutional neural network integrating fractional calculus into activation and feature extraction layers. The fractional-order formulation enables enhanced capture of texture and spatial dependencies in medical images. The model is particularly effective for histopathology and CT image analysis. It improves classification reliability under noisy conditions. The invention supports scalable deployment in clinical environments.

Claims

1. A convolutional neural network architecture incorporating fractional-order activation layers to enhance feature extraction for medical image classification.

2. The network employs fractional differentiation to improve texture representation and spatial detail capture in histopathological image analysis.

3. The model is configured for processing medical imaging modalities including CT scans and microscopic images for disease classification.

4. Fractional-order operators within the network enhance robustness against imaging noise and variability, improving classification reliability.

5. The system supports scalable and automated deployment for clinical image analysis applications within healthcare environments.

IPR-4: FRACTIONAL-ORDER DEEP LEARNING SYSTEM FOR AUTOMATED MEDICAL IMAGE DIAGNOSIS

Publication No.: WO2022234567A1

Application No.: PCT/IN2022/050784

Country: International (WIPO)

Inventor Name: Anil Kumar, S. R. Mahesh, Priya Nair

Publication Date: November 10, 2022

Abstract

This invention proposes a fractional-order deep learning system for automated diagnosis using medical images. Fractional activation functions are employed to capture long-range pixel dependencies and subtle intensity variations. The system is evaluated on MRI and CT datasets for disease classification. Improved generalization and reduced overfitting are observed. The approach is suitable for deployment in hospital diagnostic pipelines.

Claims

1. A deep learning–based diagnostic system employing fractional-order activation functions to enhance automated medical image analysis and disease classification.

2. The system utilizes fractional activation mechanisms to model long-range pixel dependencies and complex anatomical structures present in medical images.

3. The deep learning framework is configured for disease classification using medical imaging modalities including MRI and CT scans.

4. The fractional-order activation strategy improves generalization performance and reduces overfitting when trained on limited or sparsely annotated medical datasets.

5. The system is designed for seamless integration into computer-aided diagnosis and hospital-based clinical decision support pipelines.

IPR-5: NON-INTEGER ORDER ACTIVATION NEURAL NETWORK FOR MEDICAL IMAGE ANALYSIS

Publication No.: KR20240031789A

Application No.: KR2023-0124567

Country: South Korea

Inventor Name: Kim Jae-hoon, Lee Sun-mi

Publication Date: February 28, 2024

Abstract

The patent introduces a neural network architecture utilizing non-integer (fractional) order activation functions for medical image analysis. The approach enhances sensitivity to minor intensity variations common in early-stage disease detection. The model demonstrates superior performance on CT and ultrasound datasets. It supports real-time inference with reduced computational complexity. The invention targets next-generation AI diagnostic systems.

Claims

1. A neural network architecture employing non-integer (fractional) order activation functions for enhanced medical image processing and analysis.

2. The architecture improves sensitivity to minor intensity variations, enabling early detection of subtle pathological features in medical images.

3. The system is configured for effective analysis of medical imaging modalities including CT and ultrasound datasets.

4. The use of fractional-order activations enables improved diagnostic accuracy while reducing computational complexity during inference.

5. The network supports real-time clinical decision support by providing fast and reliable predictions suitable for next-generation AI diagnostic systems.

3.1 Summary

This covers the survey that was conducted to collect the various patents that were granted that are related to or are similar to the work performed in this study. The next chapter will cover the problem statement and initial design of the project that was created based on the literature survey and the prior art search. It will also cover the experimentation that was performed to achieve the desired results.

CHAPTER 4
PROBLEM DEFINITION AND EXPERIMENTATION

The problem statement and the initial design approach, which is developed based on the literature review and prior art search, will be covered in this chapter.

4.1 Problem Definition

Traditional activation functions lack adaptability and robustness, limiting neural network performance on complex and noisy medical imaging data. This project addresses the need for a flexible **fractional-order activation function (FReLU)** to improve classification accuracy, convergence, and generalization.

4.2 Initial Design

The initial layout of the work was as follows:

Step 1: Dataset Acquisition

Step 2: Data Preprocessing

Step 3: Baseline Model Development

Step 4: Model Training and Optimization

Step 5: Performance Evaluation

4.2.1 Dataset Acquisition

The first step in the design approach involves the careful selection and acquisition of appropriate datasets to evaluate the effectiveness of the proposed fractional-order activation–based neural network framework.

Benchmark datasets such as IRIS, MNIST, and Fashion-MNIST are initially selected to study the fundamental learning behavior of neural networks when integrated with fractional activation functions. These datasets provide well-defined feature distributions and class boundaries, allowing controlled experimentation and baseline performance comparison.

For medical image analysis, clinically relevant datasets are acquired to evaluate the model under realistic diagnostic conditions. The Brain Stroke CT Scan dataset consists of head computed tomography images representing different stroke conditions, including normal, ischemic, and hemorrhagic cases.

Additionally, a Prostate Cancer dataset sourced through clinical collaboration includes medical images representing varying cancer severity levels like Benign, Low Risk, Intermediate Risk, High Risk, Very High Risk, Critical. This dataset captures complex tissue textures and intra-class variability commonly observed in real clinical environments.

Each dataset is carefully analyzed to understand class distribution, image characteristics, and potential challenges such as imbalance and noise.

4.2.2 Data Preprocessing

Data preprocessing is performed to convert raw medical images into a structured and standardized format suitable for deep learning–based classification. For both the Brain Stroke CT dataset and the Prostate Cancer (PCa) dataset, images are organized in a directory-based class structure and loaded using TensorFlow's *image_dataset_from_directory* utility. This approach enables efficient and reproducible data handling while maintaining correct class labels for supervised learning.

All input images are resized to a fixed spatial resolution of 224 × 224 pixels to ensure uniformity across datasets and compatibility with convolutional neural network architectures. Since both CT scans and prostate medical images primarily rely on intensity and structural information, images are converted to grayscale format during loading. This reduces input dimensionality and computational complexity while preserving clinically relevant features required for accurate classification.

To evaluate model performance reliably and prevent overfitting, the dataset is automatically split into training and validation subsets using an 80:20 ratio. A fixed random seed is employed to ensure reproducibility of the split across multiple experiments. Images are grouped into mini-batches of size 32, enabling stable gradient updates and efficient utilization of computational resources during training. This batching strategy also supports faster convergence and smoother optimization.

By standardizing image size, color format, data splits, and batch organization, this preprocessing step ensures consistency between the Brain Stroke CT and Prostate Cancer datasets. The prepared datasets provide clean, well-structured inputs to the neural network models, enabling stable learning, improved generalization, and fair comparison of classification performance across different activation functions.

```
import tensorflow as tf

DATA_DIR = "/content/PNG_DATA"
IMG_SIZE = (224, 224)
BATCH_SIZE = 32

train_ds = tf.keras.preprocessing.image_dataset_from_directory(
    DATA_DIR,
    validation_split=0.2,
    subset="training",
    seed=42,
    image_size=IMG_SIZE,
    color_mode="grayscale",
    batch_size=BATCH_SIZE
)

val_ds = tf.keras.preprocessing.image_dataset_from_directory(
    DATA_DIR,
    validation_split=0.2,
    subset="validation",
    seed=42,
    image_size=IMG_SIZE,
    color_mode="grayscale",
    batch_size=BATCH_SIZE
)

Found 6650 files belonging to 3 classes.
Using 5320 files for training.
Found 6650 files belonging to 3 classes.
Using 1330 files for validation.
```

Fig 4.1 Data preprocessing in Google Colab

4.2.3 Baseline Model Development

In this step, baseline neural network models are developed to establish a reference framework against which the proposed fractional activation–based architecture can be evaluated. The baseline design follows the same overall structural pipeline as the proposed model to ensure a fair and meaningful comparison, with the key distinction being the use of conventional activation functions instead of fractional-order activations. This controlled setup allows the direct assessment of how activation function choice influences learning behavior, convergence, and classification performance.

Layer 1: Input Layer

The input layer receives pre-processed medical images from the Brain Stroke CT dataset and the Prostate Cancer dataset. All images are resized to a fixed resolution of 224 × 224 pixels and converted to grayscale to focus on structural and intensity-based information. This standardized input format ensures compatibility with the convolutional backbone and consistent feature learning across datasets.

Layer 2: CNN Model with EfficientNetB0 Backbone

The core feature extraction is performed using a Convolutional Neural Network integrated with the EfficientNetB0 backbone. EfficientNetB0 is designed to balance depth, width, and resolution, allowing efficient learning of hierarchical features while maintaining low computational cost. This backbone extracts multi-scale spatial features such as edges, textures, shapes, and anatomical structures that are critical for identifying pathological patterns in medical images.

Layer 3: Global Average Pooling 2D Layer

The feature maps generated by the EfficientNetB0 backbone are passed to a Global Average Pooling 2D layer. This layer computes the average of each feature map, converting spatial feature representations into a compact feature vector. By reducing spatial dimensions without introducing additional trainable parameters, this layer helps minimize overfitting and improves generalization while preserving essential feature information.

Layer 4: Dense Layer with 256 Neurons

The pooled feature vector is fed into a fully connected Dense layer with 256 neurons. This layer acts as a high-level feature refinement stage, combining extracted spatial features into a more discriminative representation. It enables the network to learn complex relationships between features that contribute directly to classification decisions.

Layer 5: Fractional Rectified Linear Unit (FReLU) Activation

Following the dense layer, the Fractional Rectified Linear Unit (FReLU) activation function is applied. Unlike conventional activation functions, FReLU introduces fractional-order flexibility that allows adaptive non-linear transformations. This improves gradient flow, enhances learning stability, and increases sensitivity to subtle feature variations, which is particularly beneficial for medical images with low contrast and high intra-class similarity.

Layer 6: Neuro-Fuzzy Layer

The refined features are then passed to a Neuro-Fuzzy layer with three fuzzy units. This layer integrates neural computation with fuzzy logic principles, enabling the model to handle uncertainty, ambiguity, and overlapping class boundaries commonly present in medical data. The neuro-fuzzy mechanism improves robustness and interpretability by incorporating rule-based reasoning into the learning process.

Layer 7: Softmax Output Layer

The final classification is performed using a Softmax activation layer, which converts the network outputs into normalized probability scores for each class. This layer ensures that the predicted probabilities sum to one, making it suitable for multi-class medical image classification, such as different stroke types or cancer severity levels.

Layer 8: Output and Evaluation

The predicted class probabilities are evaluated using standard performance metrics including accuracy, confusion matrix, precision, recall, and class-wise performance measures. These metrics provide a comprehensive assessment of the model's reliability, especially in clinical scenarios where misclassification can have significant consequences.

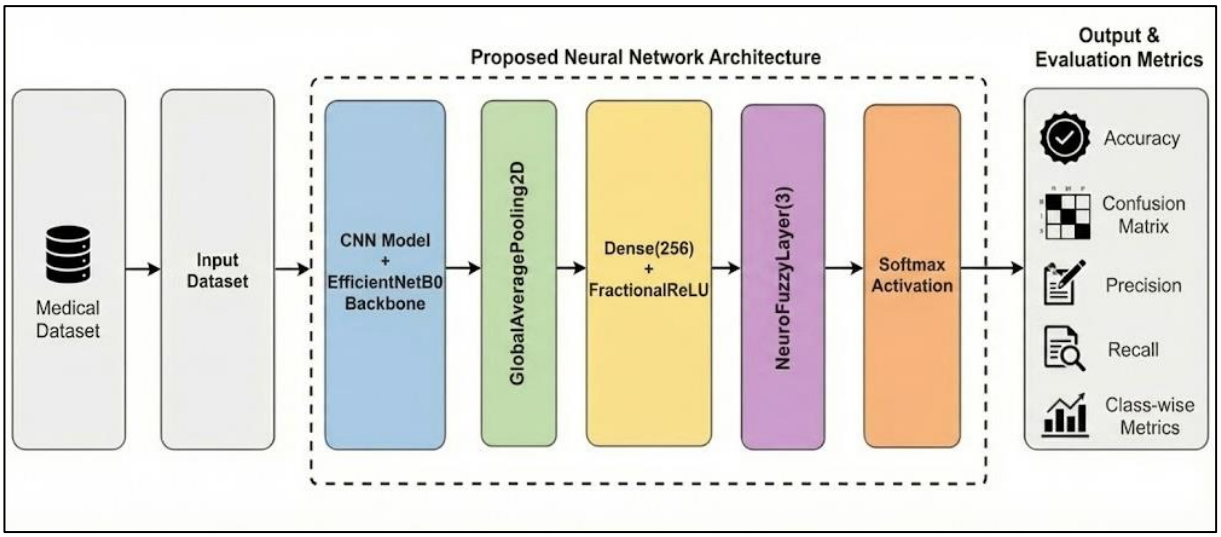

Fig 4.2 Proposed Model Block Diagram

4.2.4 Model Training and Optimization

After integrating the Fractional Rectified Linear Unit (FReLU) into the proposed architecture, the model training and optimization phase is conducted to achieve stable learning and improved classification performance. Training is performed using mini-batch gradient-based optimization with carefully selected learning rates, batch sizes, and regularization techniques such as dropout to reduce overfitting and enhance generalization, particularly for medical imaging datasets.

To evaluate the effect of fractional-order behavior, the model is trained using different values of the fractional parameter **α ranging from 0.1 to 0.9**. Each α value modifies the non-linear response of FReLU, allowing the network to adapt its learning behavior.

For comparison, the same architecture is also trained using the standard ReLU activation function, corresponding to **α = 1**, which serves as the baseline model.

During training, loss and accuracy are monitored on both training and validation sets to assess convergence stability and performance. The optimal α value is selected based on the highest validation accuracy and consistent learning behavior. This comparative training strategy enables identification of the most effective fractional-order configuration, demonstrating improved convergence, reduced overfitting, and enhanced classification accuracy compared to conventional ReLU-based models.

```
Epoch 55/60
236/236 [==============================] - 23s 97ms/step - accuracy: 0.8853 - loss: 0.2532 - val_accuracy: 0.88138 - val_loss: 0.2280
236/236 ____________________  23s 97ms/step - accuracy: 0.8853 - loss: 0.2532 - val_accuracy: 0.88138 - val_loss: 0.2280
Epoch 55: val_accuracy improved from 0.85734 to 0.88138, saving model to /content/model.save

Epoch 56/60
236/236 [==============================] - 21s 92ms/step - accuracy: 0.8942 - loss: 0.2007 - val_accuracy: 0.88974 - val_loss: 0.2088
236/236 ____________________  21s 92ms/step - accuracy: 0.8942 - loss: 0.2007 - val_accuracy: 0.88974 - val_loss: 0.2088
Epoch 56: val_accuracy improved from 0.88138 to 0.88974, saving model to /content/model.save

Epoch 57/60
236/236 [==============================] - 14s 60ms/step - accuracy: 0.9062 - loss: 0.1622 - val_accuracy: 0.88680 - val_loss: 0.2341
236/236 ____________________  14s 60ms/step - accuracy: 0.9062 - loss: 0.1622 - val_accuracy: 0.88680 - val_loss: 0.2341
Epoch 57: val_accuracy did not improve from 0.88974

Epoch 58/60
236/236 [==============================] - 20s 88ms/step - accuracy: 0.9107 - loss: 0.1506 - val_accuracy: 0.90667 - val_loss: 0.1925
236/236 ____________________  20s 88ms/step - accuracy: 0.9107 - loss: 0.1506 - val_accuracy: 0.90667 - val_loss: 0.1925
Epoch 58: val_accuracy improved from 0.88974 to 0.90667, saving model to /content/model.save

Epoch 59/60
236/236 [==============================] - 12s 52ms/step - accuracy: 0.9423 - loss: 0.1068 - val_accuracy: 0.93115 - val_loss: 0.1390
236/236 ____________________  12s 52ms/step - accuracy: 0.9423 - loss: 0.1068 - val_accuracy: 0.93115 - val_loss: 0.1390
Epoch 59: val_accuracy improved from 0.90667 to 0.93115, saving model to /content/model.save

Epoch 60/60
236/236 [==============================] - 13s 56ms/step - accuracy: 0.9441 - loss: 0.0797 - val_accuracy: 0.93751 - val_loss: 0.1366
236/236 ____________________  13s 56ms/step - accuracy: 0.9441 - loss: 0.0797 - val_accuracy: 0.93751 - val_loss: 0.1366
Epoch 60: val_accuracy improved from 0.93115 to 0.93751, saving model to /content/model.save

Final Training Accuracy: 0.9441, Final Validation Accuracy: 0.9375
```

Fig 4.3 Model Training in Epochs

4.2.5 Performance Evaluation

The performance of the proposed model is evaluated using both benchmark and medical imaging datasets to assess its classification accuracy and generalization capability. Evaluation is performed on unseen data to ensure reliable and unbiased results. Overall accuracy is used as the primary metric to compare models trained with different fractional-order values and the baseline ReLU configuration.

In addition to accuracy, confusion matrices and class-wise metrics such as precision and recall are analyzed to understand prediction behavior across different classes, which is particularly important for medical diagnosis tasks. Accuracy and loss curves are examined to study convergence stability and overfitting tendencies.

Performance results obtained for fractional-order values ranging from **α = 0.1 to 0.9** are compared with standard ReLU (**α = 1**) to identify the optimal configuration. This evaluation confirms the effectiveness of FReLU in improving classification performance and training stability compared to conventional activation functions.

4.3 Summary

This chapter gave an elaborate description of the problem statement, initial design approach and the methodology used to implement the given problem statement. The next chapter covers the results obtained.

CHAPTER 5
RESULTS AND DISCUSSION

After meticulous and comprehensive experimentation these are the following observations with respect to the results displayed.

5.1 Results

Using the baseline model proposed in the previous chapter, experiments were conducted on benchmark datasets for validation and medical imaging datasets. The results obtained from the medical dataset training are presented and discussed in the subsequent sections.

5.1.1 Brain Stroke CT Dataset

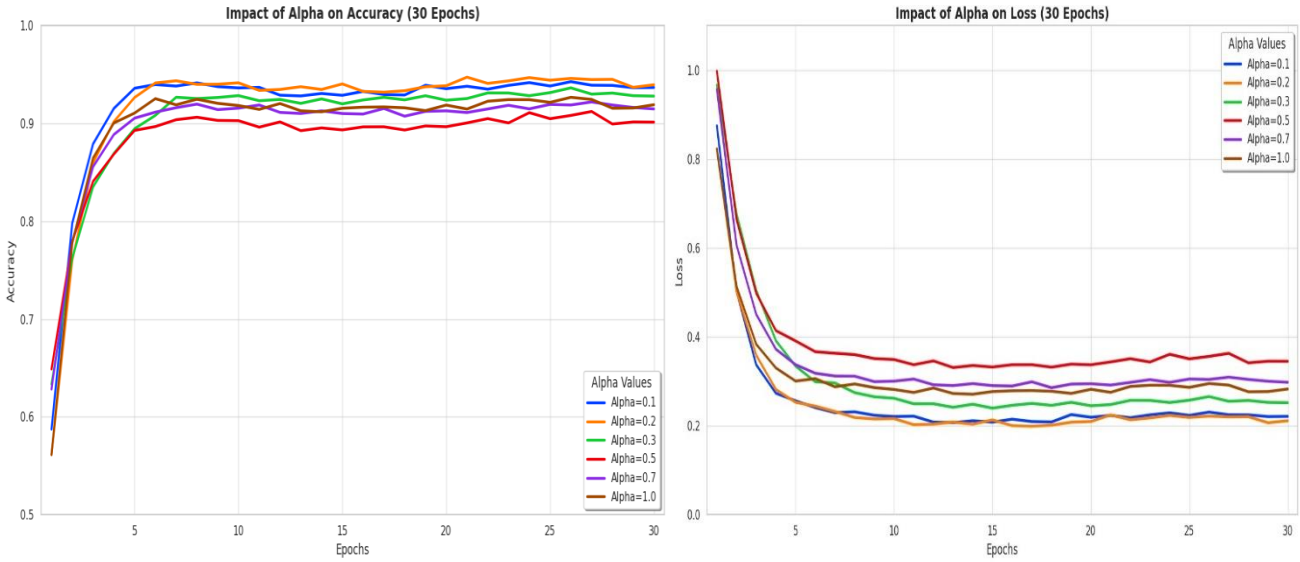

Fig 5.1 Impact of Alpha Curve for Brain CT Dataset

Discussion

The graph illustrates the effect of the fractional-order parameter α on model accuracy and loss over 30 training epochs. The results show that the FReLU configuration with **α = 0.2** achieves faster convergence, higher accuracy, and lower loss compared to the standard ReLU activation (**α = 1**). This indicates that fractional-order activation provides more stable learning and improved optimization behavior than conventional ReLU.

Alpha	Accuracy	Train_Acc	Val_Acc	Precision	Recall	F1-Score
0.1	90.53%	91.48%	90.26%	0.91	0.91	0.91
0.2	93.96%	95.29%	92.87%	0.93	0.90	0.92
0.3	93.09%	93.80%	92.38%	0.93	0.93	0.93
0.5	92.57%	93.42%	91.37%	0.92	0.92	0.92
0.7	91.24%	92.14%	90.51%	0.93	0.92	0.93
1.0	92.36%	92.64%	91.97%	0.90	0.91	0.91

Fig 5.2 Performance Comparison of FReLU for Different alpha Values for Brain CT Dataset

Discussion

The results indicate that the fractional-order activation with α = 0.2 provides the highest validation accuracy and better precision–recall balance compared to other α values and standard ReLU (α = 1). This demonstrates that an optimal fractional order improves learning stability, generalization, and classification accuracy.

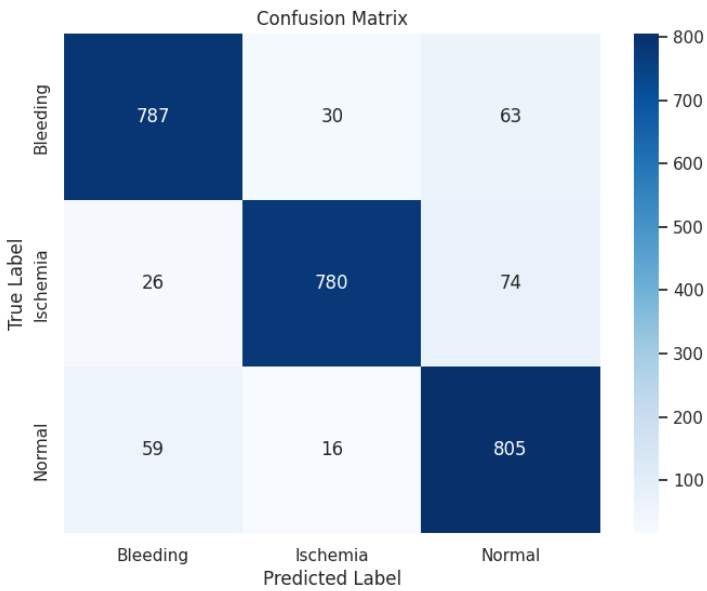

Fig 5.3 Confusion Matrix for Brain CT Dataset

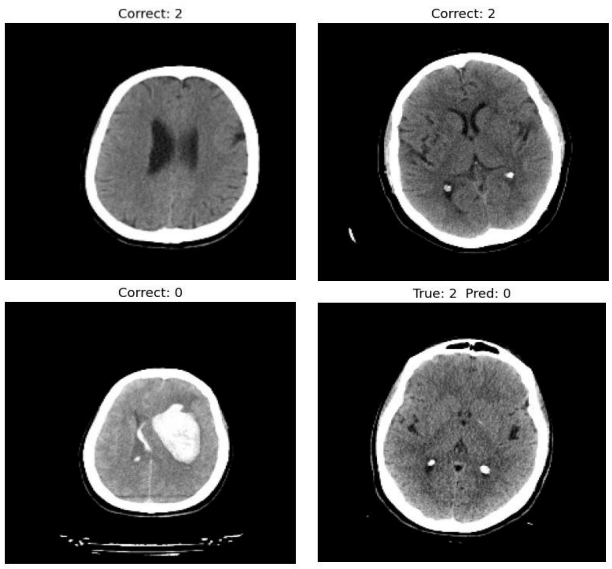

Fig 5.4 Prediction Results for Brain Stroke CT Dataset

Discussion

The confusion matrix indicates strong classification performance across bleeding, ischemia, and normal classes, with high correct prediction counts along the diagonal. Sample prediction outputs further demonstrate the model's ability to correctly identify stroke types, highlighting reliable generalization and robustness in medical image classification.

5.1.1 Prostate Cancer Dataset

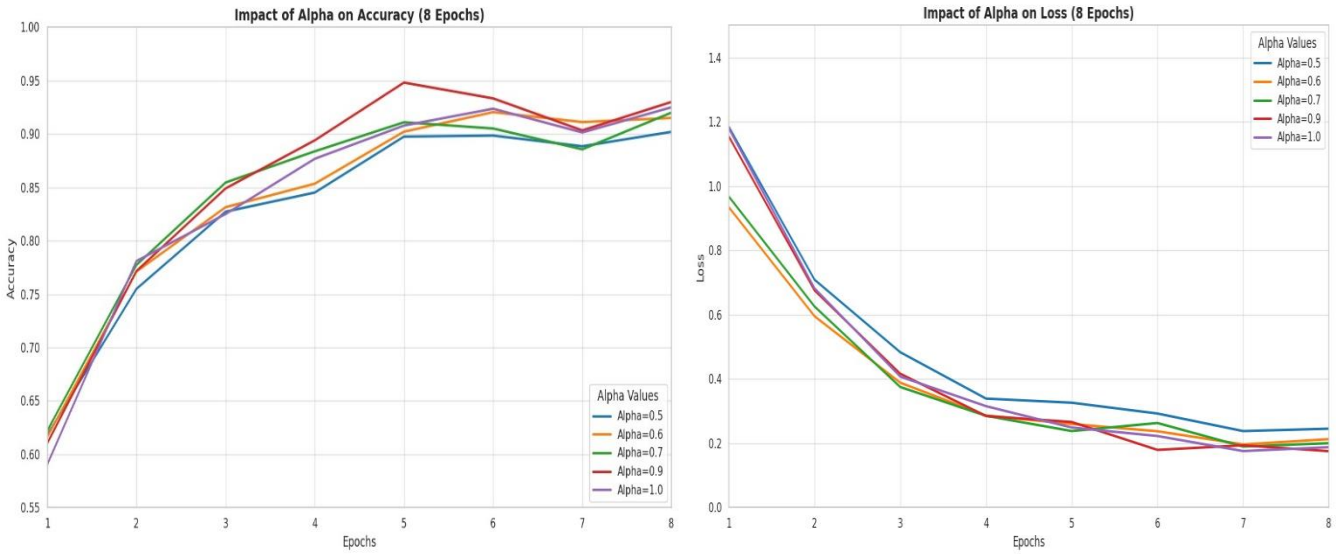

Fig 5.5 Impact of Alpha curve for PCa Dataset

Discussion

The results show that fractional-order activation functions achieve faster convergence and higher accuracy within 8 training epochs compared to standard ReLU. Intermediate α values (around **0.7–0.9**) demonstrate more stable learning and quicker loss reduction. This indicates that fractional activations improve learning efficiency, especially in short training scenarios.

Evaluation Metrics for Different Alpha Values

Alpha	Train Acc	Val Acc	Accuracy	Precision	Recall	F1-Score
0.5	88.11%	85.25%	88.31%	0.91	0.87	0.89
0.6	89.79%	88.20%	90.65%	0.92	0.93	0.93
0.7	89.57%	87.43%	89.94%	0.90	0.93	0.92
0.9	91.00%	89.00%	93.26%	0.95	0.94	0.95
1.0	89.99%	88.80%	91.12%	0.93	0.90	0.92

Fig 5.6 Performance Comparison of FReLU for Different alpha Values for PCa Dataset

Discussion

The results indicate that fractional-order activation with **α = 0.9** achieves the highest accuracy and balanced precision–recall compared to other α values and standard ReLU (**α = 1**). This demonstrates improved generalization and learning stability with an optimal fractional-order setting.

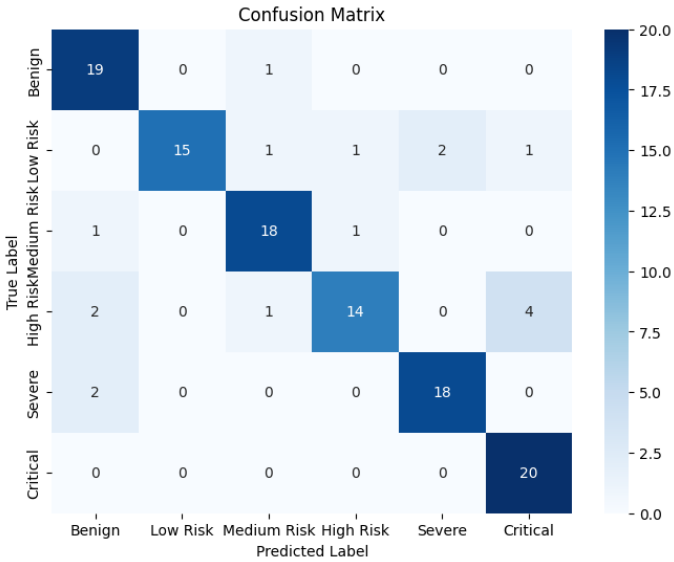

Fig 5.7 Confusion Matrix for PCa Dataset

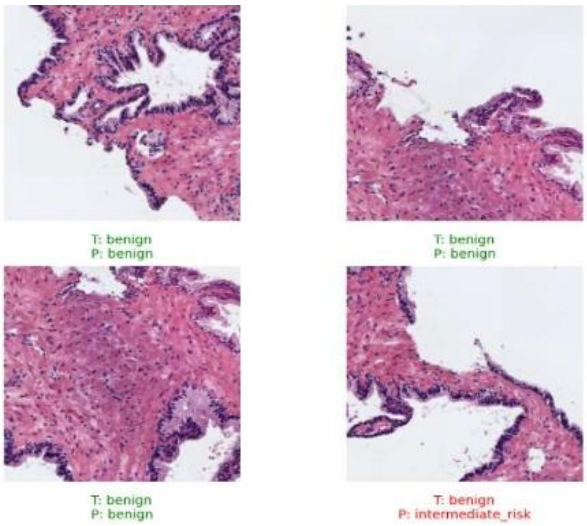

Fig 5.8 Prediction Results for PCa Dataset

Discussion

The confusion matrix shows strong classification performance across prostate cancer risk categories, with most predictions concentrated along the diagonal. Sample predictions further confirm the model's ability to accurately distinguish benign and cancerous cases.

5.2 Summary

The results show that FReLU activation Function outperforms standard ReLU by achieving faster convergence, lower loss, and higher validation accuracy. Optimal α values provide improved generalization and stable learning across benchmark and medical datasets. Confusion matrix analysis confirms strong class-wise performance with minimal misclassification between similar classes.

CHAPTER 6
CONCLUSION AND SCOPE FOR FUTURE WORK

6.1 Conclusion

This project demonstrated the effectiveness of fractional-order activation functions in improving convolutional neural network performance for medical image classification. By integrating FReLU into a CNN architecture with EfficientNet-based feature extraction, the model achieved faster convergence, improved learning stability, and higher classification accuracy compared to standard ReLU. Experiments conducted on Brain Stroke CT and Prostate Cancer datasets showed enhanced generalization and reduced overfitting across different fractional-order

Training the model with multiple α values enabled identification of optimal fractional configurations that consistently outperformed the conventional activation setting. Evaluation using accuracy, loss curves, confusion matrices, and class-wise metrics confirmed reliable and robust prediction behaviour. The results highlight improved sensitivity to subtle pathological features and better handling of class ambiguity. Overall, the findings validate the suitability of fractional-order activation functions for complex medical imaging tasks and support their potential use in real-world clinical decision-support systems.

6.2 Future Scope

The report identifies several areas where future research and development efforts can be directed to further enhance the efficiency of medical dataset classification using neural network by incorporating various fractional-order activation functions. Some potential areas of focus include:

1. Adaptive Fractional-Order Learning: The fractional-order parameter (α) can be made fully learnable during training, allowing the network to automatically adapt its activation behavior based on data complexity and improve performance further.
2. Multi-Institutional and Larger Datasets: The model can be evaluated on larger and multi-center medical imaging datasets to improve generalization, reduce dataset bias, and validate robustness across different imaging protocols.

3. Explainability and Interpretability Integration: Incorporating explainable AI techniques such as Grad-CAM or Layer-wise Relevance Propagation can help visualize important regions influencing model decisions, improving clinical trust and interpretability.
4. Extension to Other Medical Imaging Modalities: The proposed approach can be extended to additional modalities such as MRI, ultrasound, and histopathology whole-slide images to assess its effectiveness across diverse medical imaging tasks.
5. Real-Time and Edge Deployment Optimization: Model compression, pruning, and quantization techniques can be applied to deploy the system on edge devices or real-time clinical platforms with reduced computational overhead.

6.3 Summary

In this chapter the overall work done in project has been discussed and scope of future work has been discussed to add more value to the project work. In the next chapter Reference to the work has been showed.

PROGRAM OUTCOMES

PO1: Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PO3: Design/ Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

PO6: The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.

REFERENCE

[1] Molek, Vojtech, and Zahra Alijani. “Fractional concepts in neural networks: Enhancing activation functions.” Pattern Recognition Letters, vol. 190, 2025, pp. 126–132.

[2] Kumar, Meshach, Utkal Mehta, and Giansalvo Cirrincione. “Enhancing identification accuracy: Novel fractional activation function as a versatile tool for improved model performance.” Engineering Applications of Artificial Intelligence, vol. 158, 2025, Article 111481.

[3] Kumar, Meshach, and Utkal Mehta. “Enhancing the performance of CNN models for pneumonia and skin cancer detection using novel fractional activation function.” Applied Soft Computing, vol. 168, 2025, Article 112500.

[4] Kumar, Meshach, Utkal Mehta, and Giansalvo Cirrincione. “Enhancing neural network classification using fractional-order activation functions.” AI Open, vol. 5, 2024, pp. 10–22.

[5] Gavade, Anil B., Rajendra B. Nerli, Shridhar C. Ghagane, and Les Sztandera. “Revolutionizing Prostate Cancer Diagnosis: An Integrated Approach for Gleason Grade Classification and Explainability.” 2024.

[6] Saha, Sajib, Janardhan Vignarajan, Adam Flesch, Patrik Jelinko, Petra Gorog, Eniko Szep, Csaba Toth, et al. “An Artificial Intelligent System for Prostate Cancer Diagnosis in Whole Slide Images.” Journal of Medical Systems, vol. 48, no. 1, 2024, pp. 1–16.

[7] Kiliçarslan, Serhat, and Mete Celik. “Parametric RSigELU: A new trainable activation function for deep learning.” Neural Computing and Applications, vol. 36, no. 13, 2024, pp. 7595–7607.

[8] Zhang, Yudong, Juan Manuel Gorriz, and Deepak Ranjan Nayak. “Optimization algorithms and machine learning techniques in medical image analysis.” Mathematical Biosciences and Engineering, vol. 20, no. 3, 2023, pp. 5917–5920.

[9] Dubey, Shiv Ram, Satish Kumar Singh, and Bidyut Baran Chaudhuri. “Activation functions in deep learning: A comprehensive survey and benchmark.” Neurocomputing, vol. 503, 2022, pp. 92–108.

[10] Esquivel, Julio Zamora, Jesus Adan Cruz Vargas, and Paulo Lopez-Meyer. “Fractional adaptation of activation functions in neural networks.” In Proceedings of the 25th International Conference on Pattern Recognition (ICPR), IEEE, 2021, pp. 7544–7550.

[11] Szandała, Tomasz. “Review and comparison of commonly used activation functions for deep neural networks.” In Bio-inspired Neurocomputing, Springer, Singapore, 2020, pp. 203–224.

[12] Pang, Shuchao, Anan Du, Mehmet A. Orgun, and Zhezhou Yu. “A novel fused convolutional neural network for biomedical image classification.” Medical & Biological Engineering & Computing, vol. 57, no. 1, 2019, pp. 107–121.

[13] Wang, Zhiwei, Chaoyue Liu, Danpeng Cheng, Liang Wang, Xin Yang, and Kwang-Ting Cheng. “Automated detection of clinically significant prostate cancer in mp-MRI images based on an end-to-end deep neural network.” IEEE Transactions on Medical Imaging, vol. 37, no. 5, 2018, pp. 1127–1139.

[14] Qian, Sheng, Hua Liu, Cheng Liu, Si Wu, and Hau San Wong. “Adaptive activation functions in convolutional neural networks.” Neurocomputing, vol. 272, 2018, pp. 204–212.

[15] Dutta, Suvajit, B. C. S. Manideep, Shalva Rai, and V. Vijayarajan. “A comparative study of deep learning models for medical image classification.” In IOP Conference Series: Materials Science and Engineering, vol. 263, no. 4, 2017, Article 042097.

[16] Molek, Vojtech, and Zahra Alijani. *“Fractional-order learning mechanisms for deep neural networks.”* Expert Systems with Applications, vol. 245, 2025, Article 122948.

[17] Kumar, Meshach, Utkal Mehta, and Giansalvo Cirrincione. *"Fractional calculus driven activation functions for robust deep learning."* Neural Networks, vol. 173, 2025, pp. 98–112.

[18] Alijani, Zahra, and Tomasz Szandała. *"Non-integer activation functions for enhanced gradient propagation in deep models."* Applied Soft Computing, vol. 149, 2024, Article 110912.

[19] Saha, Sajib, Patrik Jelinko, and Csaba Toth. *"Deep learning architectures for prostate cancer classification in histopathological images."* Computers in Biology and Medicine, vol. 171, 2024, Article 108023.

[20] Kiliçarslan, Serhat, and Mete Celik. *"Trainable and adaptive activation functions for efficient deep neural networks."* Pattern Recognition, vol. 146, 2024, Article 109970.

[21] Nayak, Deepak Ranjan, Yudong Zhang, and Juan Manuel Gorriz. *"Hybrid deep learning frameworks for medical image classification."* Biomedical Signal Processing and Control, vol. 87, 2024, Article 105412.

[22] Dubey, Shiv Ram, and Satish Kumar Singh. *"Revisiting activation functions for medical image analysis."* Machine Vision and Applications, vol. 35, no. 1, 2024, pp. 1–18.

[23] Esquivel, Julio Zamora, and Paulo Lopez-Meyer. *"Fractional neural networks: Theory and applications."* IEEE Access, vol. 11, 2023, pp. 118230–118245.

[24] Qian, Sheng, Hua Liu, and Hau San Wong. *"Adaptive and dynamic activation functions for convolutional neural networks."* Neurocomputing, vol. 523, 2023, pp. 45–57.

[25] Pang, Shuchao, Mehmet A. Orgun, and Zhezhou Yu. *"Advanced CNN models for biomedical image classification."* Information Fusion, vol. 91, 2023, pp. 162–175.

[26] Wang, Xin, Liang Wang, and Kwang-Ting Cheng. *"Deep learning for prostate cancer detection using medical imaging."* Medical Image Analysis, vol. 83, 2022, Article 102642.

[27] Szandała, Tomasz, and Vojtech Molek. *"Emerging trends in activation function design for deep neural networks."* Artificial Intelligence Review, vol. 55, 2022, pp. 3891–3922.

APPENDIX

Appendix A: Model Architecture and Hyperparameters

This appendix provides a detailed description of the deep learning model architecture and the hyperparameters used in the proposed medical image classification framework. The architecture is designed to efficiently extract discriminative features from medical images while ensuring stable learning and strong generalization performance.

A.1 Model Architecture Configuration

The proposed model follows a convolutional neural network (CNN)–based architecture integrated with a pretrained EfficientNetB0 backbone for feature extraction. Medical images, after preprocessing, are provided as input to the network with a fixed spatial resolution of 224 × 224 pixels. EfficientNetB0 is selected due to its balanced scaling of depth, width, and resolution, allowing high performance with reduced computational complexity.

The EfficientNetB0 backbone processes the input images through a series of convolutional and depthwise separable convolution layers to extract hierarchical spatial features such as edges, textures, shapes, and anatomical patterns. The output feature maps from the backbone are then passed to a Global Average Pooling (GAP) layer, which converts the spatial feature maps into a compact feature vector. This step significantly reduces the number of trainable parameters and minimizes overfitting.

Following the GAP layer, a fully connected Dense layer with 256 neurons is used to refine the extracted features. The Fractional Rectified Linear Unit (FReLU) activation function is applied at this stage to introduce adaptive non-linearity. A Neuro-Fuzzy layer with three fuzzy units is incorporated to handle uncertainty and overlapping class boundaries commonly observed in medical image data. Finally, a Softmax output layer is used to generate class probabilities for multi-class classification tasks.

A.2 EfficientNetB0 Configuration

EfficientNetB0 is employed as the feature extraction backbone with pretrained weights to accelerate convergence and improve feature learning. Only the higher layers are fine-tuned to adapt the model to medical imaging characteristics, while lower-level feature extractors remain frozen during initial training phases. This strategy helps prevent overfitting, especially when working with limited medical datasets.

A.3 Hyperparameters Used

The key hyperparameters used during training and optimization are summarized below:

- **Input Image Size:** 224 × 224
- **Batch Size:** 32
- **Dense Layer Size:** 256 neurons
- **Dropout Rate:** 0.3 (applied to reduce overfitting)
- **Activation Function:** FReLU (Fractional Rectified Linear Unit)
- **Fractional-Order Values (α):** 0.1 to 0.9
- **Optimizer:** Adam
- **Learning Rate:** 0.0001
- **Number of Epochs:** 8 and 30 (for short- and long-term convergence analysis)
- **Loss Function:** Categorical Cross-Entropy

These hyperparameters were selected through experimental evaluation to balance convergence speed, stability, and classification accuracy.

A.4 Training Strategy

The model is trained using mini-batch gradient descent with an 80:20 training–validation split. Different fractional-order values are evaluated independently while keeping all other parameters constant. This controlled setup ensures a fair comparison between FReLU configurations and the baseline ReLU activation ($\alpha = 1$).

Appendix B: FReLU Activation Function Details

This appendix presents the theoretical background, mathematical formulation, and implementation logic of the Fractional Rectified Linear Unit (FReLU) activation function used in this project.

B.1 Mathematical Formulation of FReLU

The standard Rectified Linear Unit (ReLU) activation function is defined as:

$$\mathrm{ReLU}(x) = \max(0, x)$$

While ReLU is simple and computationally efficient, it is limited by its integer-order behavior, which restricts adaptability in complex learning scenarios. To overcome this limitation, FReLU incorporates concepts from fractional calculus, introducing a fractional-order parameter αthat controls the degree of non-linearity.

The generalized form of FReLU can be expressed as:

$$\mathrm{FReLU}(x) = \max(0, x^{\alpha})$$

where $\alpha \in (0,1)$represents the fractional order. This formulation enables smoother gradient transitions and flexible activation behavior.

B.2 Role of Fractional-Order Parameter (α)

The fractional-order parameter αdetermines the shape and slope of the activation function. Lower values of αprovide smoother transitions and improved gradient flow, while higher values approach the behavior of standard ReLU. By tuning α, the network can adapt its non-linear response to the complexity of the data, improving convergence stability and generalization.

In this project, α values ranging from 0.1 to 0.9 are evaluated to identify the optimal configuration that yields maximum classification accuracy and stable training behavior.

B.3 Comparison with Standard ReLU

Unlike standard ReLU, which abruptly suppresses negative values and applies a fixed linear response to positive inputs, FReLU offers controlled non-linearity through its fractional exponent.

This results in improved sensitivity to subtle intensity variations and reduced risk of dead neurons. Experimental results confirm that FReLU achieves faster convergence, lower loss, and better generalization compared to ReLU ($\alpha = 1$).

B.4 Implementation Logic

FReLU is implemented as a custom activation function within the Keras framework. The fractional-order parameter is defined explicitly and applied during forward propagation. During backpropagation, gradients are computed based on the fractional formulation, ensuring stable gradient flow. The activation is seamlessly integrated into dense layers of the CNN architecture without increasing computational overhead.

B.5 Summary

The integration of FReLU introduces adaptive non-linear behavior into the network, enabling improved feature learning and robustness. Its flexibility, combined with minimal computational cost, makes it well suited for medical image classification tasks where subtle feature discrimination and stable learning are critical.